

Headline results:

$$\tilde{z} = m[\lambda_a^2 + \lambda_b^2] + [g\mu_a + (1-g)\mu_b]^{-1}$$

$$\lambda_a = hg + (1-g)(1-h)$$

$$\lambda_b = h(1-g) + (1-h)g$$

$$\pi_a = \frac{\lambda_a \tilde{z} K^{(in)}}{mt}$$

$$\text{prob("a" node has in degree} = k^{(in)}) = p(k^{(in)}) = A_a \frac{\Gamma(k^{(in)} + \mu_{\lambda_a} + 1 + \frac{z_{\lambda_a}}{2})}{\Gamma(k^{(in)} + \mu_{\lambda_a})} \quad \hookrightarrow \text{theoretical "a" degree distribution}$$

$$A_a = \frac{\Gamma(\mu_{\lambda_a})}{\Gamma(\mu_{\lambda_a} + 1 + \frac{z_{\lambda_a}}{2})} \cdot (1 + \mu_a \tilde{z}^{-1})$$

$$\pi_b = \frac{\lambda_b \tilde{z} K^{(in)}}{mt}$$

$$\text{prob("b" node has in degree} = k^{(in)}) = p(k^{(in)}) = A_b \frac{\Gamma(k_b + \mu_{\lambda_b} + 1 + \frac{z_{\lambda_b}}{2})}{\Gamma(k_b + \mu_{\lambda_b})} \quad \hookrightarrow \text{theoretical "b" degree distribution}$$

$$A_b = \frac{\Gamma(\mu_{\lambda_b})}{\Gamma(\mu_{\lambda_b} + 1 + \frac{z_{\lambda_b}}{2})} \cdot (1 + \mu_b \tilde{z}^{-1})$$

$$\left. \begin{array}{l} \text{initialise with: } N(t_{\text{init}}) = t_{\text{init}} \\ \quad \quad \quad E(t_{\text{init}}) = mt_{\text{init}} \end{array} \right\} \rightarrow (\text{Like in Tim-tau})$$